

Unit 13, Lesson 2: Evaluating Algebraic Expressions Using Substitution



Benchmark

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.



Learning Target

☐ I can evaluate algebraic expressions using substitution.



13.2.1 Warm-Up: Order Explorer

1. Place a digit in each box to create an expression with the largest possible value. Use any of the digits 1 to 5, once each at most.

$$\begin{array}{c} \square \\ \square + \square \times \square \end{array}$$

2. What is the value of your expression?
3. On a scale of 1 to 10, how confident are you that your expression is the largest possible value?
4. Explain your strategy for choosing and placing the numbers as you did.



13.2.2 Collaboration: Evaluating Algebraic Expressions by Substituting Integers

1. Brae evaluated the expression $7b + 12 - (5 - 3)^2$ for $b = -2$. Her work is shown.
 - a. Evaluate her work and determine where she made mistakes.

$$\begin{array}{r} 7b + 12 - (5 - 3)^2 \\ 7 - 2 + 12 - (5 - 3)^2 \\ 7 - 2 + 12 - (-2)^2 \\ 7 - 2 + 12 + 4 \\ 5 + 12 + 4 \\ 17 + 4 \\ 21 \end{array}$$

- b. Help Brae understand and learn from her mistakes. Write a brief note to Brae explaining what her mistakes were and how she could learn from the mistakes.

Means
I
Start
To
Acquire
Knowledge
Experience
Skills

- c. Read the note your partner wrote and offer suggestions on ways to improve their note.
 - d. Using the word bank below, complete the three reminders that could help Brae.

inside outside parentheses

- When substituting values into algebraic expressions, it is helpful to put the substituted value(s) in _____.
 - When evaluating expressions with exponents, if the negative is _____ the parentheses, the negative indicates subtraction, or the opposite of the value found by raising the value inside to a power.
 - If the negative is _____ the parentheses, it indicates that the number is a negative number being raised to a power.
- e. Verify your answers with your partner and, if necessary, resolve any differences.

2. Complete the following problems.

a. Evaluate the expression $-8m - 4q^2 + \frac{s}{3}$ where $m = 4$, $q = -3$, and $s = -24$.

b. Evaluate the expression $3s - 4q - m^2$ where $m = 4$, $q = -3$, and $s = -24$.

c. Evaluate the expression $-m^3 - 3q + (s - q)$ where $m = 4$, $q = -3$, and $s = -24$.

d. All of the expressions should have the same result. Check your work with your partner.



13.2.3 Your Turn!

1. What is the value of $5(6x + 7) - 3y^2$ if $x = -1$ and $y = 5$?

2. Evaluate the expression $3d^4 - \frac{w}{3}$ where $d = 4$ and $w = -75$.

3. Find the value of $-2(b + 4) + 3a - (7 - a)^5$ if $a = -3$ and $b = 5$.

4. Evaluate the expression $c^3 - 5k + (h - k)$ where $c = -9$, $h = 29$, and $k = -17$.

5. Evaluate each expression.

a. $8 \div p - 16 \div (-5 + 7)$ where $p = 4$

b. $13 + \frac{6}{y} - y^3$ if $y = -3$

c. $4g - 12 + (h - 7) \cdot 3^2$ where $g = -7$ and $h = 6$

d. $w \div t \cdot H \div y^2$ where $H = -3$, $t = 7$, $w = 84$, and $y = -2$



13.2.4 Wrap-Up: Reflection

1. Color in the arrow up to the statement that best describes your current understanding.



I'm so confident, I can explain evaluating algebraic expressions using substitution to someone else!

I can get to the right answer, but I don't understand well enough to explain it yet.

I understand some of this, but I don't understand all of it yet. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute a value

I tried hard, but I am finding this challenging. I checked what I do not understand yet.

- ☐ Working with negative numbers
- ☐ Finding values raised to an exponent
- ☐ Remembering to multiply and divide from left to right
- ☐ Remembering to use parentheses when I substitute in a value